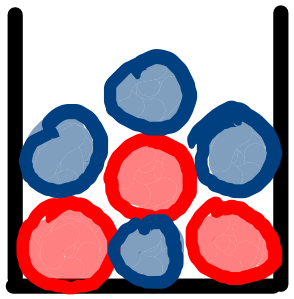
Ejemplo 1:



+ Se toman n bolillas al azar $\boxed{\text{con}}$ reposición

$X = \#$ bolillas rojas

Ejemplo 2:

+ Se tira n veces la moneda

$X = \#$ de veces que sale cara

1. X = cuenta "éxitos"
2. Se repite un experimento n veces idénticas, es decir, la probabilidad de "éxito" no cambia
3. Los experimentos son independientes

Ejemplo 1: $n = 5$, p = probabilidad de "éxito" $= 3/7$

$$R_x = \{0, 1, 2, 3, 4, 5\} = \{0, 1, \dots, n\}$$

$$P(X=0) = (1-p)^5 = q^5$$

$$\underline{A}\ \underline{A}\ \underline{A}\ \underline{A}\ \underline{A} \rightarrow (1-p)\cdot(1-p)\cdot(1-p)\cdot(1-p)\cdot(1-p)$$

$$= (\underbrace{1-p}_{=q})^5 = q^5$$

$$P(X=1) = 5 \cdot p \cdot q^4 = \binom{5}{1} \cdot p^1 \cdot q^4$$

$$\left.\begin{array}{lcccccc}
p \cdot q^4 \leftarrow & \underline{R} & \underline{A} & \underline{A} & \underline{A} & \underline{A} \\
q \cdot p \cdot q^3 \leftarrow & A & R & A & A & A \\
p\, q^4 \leftarrow & A & A & R & A & A \\
p\, q^4 \leftarrow & A & A & A & R & A \\
p\, q^4 \leftarrow & A & A & A & A & R
\end{array}\right\} 5 \cdot p \cdot q^4$$

$$P(X=2) = 10 \cdot p^2 \cdot q^3 = \binom{5}{2} \cdot p^2 \cdot q^3$$

$p^2 \cdot q^3 \leftarrow$

R	R	A	A	A
R	A	R	A	A
R	A	A	R	A
R	A	A	A	R
A	A	A	R	R

5

A	R	R	A	A
A	R	A	R	A
A	R	A	A	R
A	A	R	R	A
A	A	R	A	R

5

10 czsos

$$P(X=0) = \binom{5}{0} p^0 q^5$$

$$P(X=1) = \binom{5}{1} \cdot p^1 q^4$$

$$P(X=2) = \binom{5}{2} \cdot p^2 \cdot q^3$$

$$\vdots$$

$$\rightarrow P(X=k) = \binom{5}{k} \cdot p^k \cdot q^{5-k}$$

$$\sum_{k \in R_X} P(X=k) = \underbrace{\sum_{k=0}^{5} \binom{5}{k} \cdot p^k \cdot q^{5-k}}_{\text{desarrollo del binomio } (p+q)^5} = (p+q)^5$$

$$= (\cancel{p} + 1 - \cancel{p})^5 = 1^5 = 1$$

$$X \sim \text{Binomial}(5, p)$$

$X \sim \text{Binomial}(n, p)$

X = número de "éxitos" entre n experimentos independientes, cada uno con la misma probabilidad de "éxito" p

+ $R_X = \{0, 1, \dots, n\}$
+ $P(X = k) = \binom{n}{k} p^k q^{n-k} \quad \forall k \in R_X$, con $q = 1 - p$
+ $\mu_X = np$
+ $\sigma_X^2 = np(1-p) = npq$

$$E[X] = \sum_{k=0}^{n} k \cdot P(X=k) = \sum_{k=0}^{n} k \cdot \binom{n}{k} p^k q^{n-k} =$$

$$= \sum_{k=1}^{n} \cancel{k} \cdot \frac{n!}{\underbrace{\cancel{k!}}_{(k-1)!}\,(n-k)!} p^k q^{n-k} =$$

$$= np \sum_{k=1}^{n} \frac{(n-1)!}{(k-1)!\,[(n-1)-(k-1)]!} p^{k-1} q^{[(n-1)-(k-1)]}$$

$$= np \sum_{m=0}^{n-1} \binom{n-1}{m} \cdot p^m \cdot q^{n-1-m} = np \cdot (p+q)^{n-1} = np = \mu_x$$

$m = k-1$

$$E[X^2] = \sum_{k=\cancel{0}\,1}^{n} k^2 \binom{n}{k} \cdot p^k q^{n-k} = \sum_{k=1}^{n} \left(k(k-1)+k\right) \cdot \binom{n}{k} \cdot p^k \cdot q^{n-k}$$

$$= \underbrace{\sum_{k=1}^{n} k(k-1) \cdot \binom{n}{k} \cdot p^k \cdot q^{n-k}}_{= \text{ver próxima página}} + \underbrace{\sum_{k=1}^{n} k \binom{n}{k} p^k q^{n-k}}_{= np}$$

$$= n(n-1)p^2 + np = n^2p^2 - np^2 + np =$$

$$= n^2p^2 + np(1-p) = n^2p^2 + npq$$

$$\sum_{k=\cancel{1}\,2}^{n} \cancel{k(k-1)} \frac{n!}{\cancel{k!}\,(n-k)!} p^k q^{n-k} =$$

$(k-2)!$

$$= n(n-1)p^2 \sum_{k=2}^{n} \frac{(n-2)!}{(k-2)!\,[(n-2)-(k-2)]!} \cdot p^{k-2} \cdot q^{[(n-2)-(k-2)]}$$

$m = k-2$

$$= n(n-1)p^2 \sum_{m=0}^{n-2} \binom{n-2}{m} \cdot p^m \cdot q^{n-2-m} = n(n-1)\cdot p^2 (p+q)^{n-2}$$

$$= n(n-1)\cdot p^2$$

$$Var(X) = E[X^2] - (E[X])^2$$

$$= \cancel{n^2p^2} + npq - \cancel{(np)^2} = npq = \sigma_X^2$$